

BST Vol 7 Ch 2 — Maxwell Equations from D_IV⁵ Substrate (v0.4.1, substantive 3-level pedagogy upgrade)

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Vol 7 Chapter 2 — Maxwell Equations from D_IV⁵ Substrate

Headline result

Maxwell's 4 equations governing electromagnetism:

Gauss's law (electric): $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$

Gauss's law (magnetic): $\nabla \cdot \mathbf{B} = 0$

Faraday's law: $\nabla \times \mathbf{E} + \partial \mathbf{B}/\partial t = 0$

Ampère-Maxwell law: $\nabla \times \mathbf{B} - (1/c^2)\partial \mathbf{E}/\partial t = \mu_0 \mathbf{J}$

These 4 equations are derived from BST substrate via T2477 (Lyra Saturday SP-31 #286): the gauge field A_μ is a connection 1-form on Bergman line bundle $L_\lambda \rightarrow D_{IV}^5$ with structure group $K = SO(5) \times SO(2)$. The field strength tensor $F_{\mu\nu}$ is the curvature of this connection.

Source equations: from Yang-Mills action variation $\delta S/\delta A = 0$.

Homogeneous equations: from Bianchi identity $dF = 0$ (geometric identity for any curvature 2-form).

Substrate mechanism

Connection 1-form on Bergman bundle (T2477):

The substrate D_{IV}^5 has Bergman line bundle $L_\lambda \rightarrow D_{IV}^5$ with structure group $K = SO(5) \times SO(2)$. $U(1)_{em}$ is a sub-bundle. The gauge potential A_μ is a connection 1-form on L_λ — a substrate-natural geometric object.

Field strength as curvature:

$$F = dA + A \wedge A$$

For abelian U(1)_em the wedge term vanishes, leaving $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

Yang-Mills action:

$$S_{YM} = -\frac{1}{4} \int_{\mu\nu} F^{\mu\nu} d^4x$$

Action variation $\delta S / \delta A_\mu = 0$ yields:

$$\partial_\mu F^{\mu\nu} = J^\nu$$

In 3D form this gives the source Maxwell equations (Gauss-E + Ampère-Maxwell).

Bianchi identity:

For any curvature 2-form $F = dA$: $dF = ddA = 0$. In components:

$$\partial_{[\mu} F_{\nu\rho]} = 0$$

This gives the source-free Maxwell equations (Gauss-B + Faraday).

Match precision

D-tier on Maxwell from substrate (T2477 RIGOROUSLY VERIFIED Saturday).
Standard EM phenomenology preserved at any precision. QED corrections
handled by Vol 2 Ch 8 (α universal α -analog) + Vol 7 Ch 7 (Relativistic EM).

Cross-volume dependencies

- Vol 1 Ch 8 (Gauge Theory + T2477) — substrate gauge field framework
- Vol 7 Ch 3 (Electrostatics) — static-electric sector of Maxwell
- Vol 7 Ch 4 (Magnetostatics) — static-magnetic sector
- Vol 7 Ch 5 (EM Waves) — wave solutions of Maxwell
- Vol 7 Ch 8 (Lagrangian EM) — variational framework underlying Yang-Mills action
- Vol 2 Ch 8 (Coupling Constants) — α universal α -analog (electroweak embedding)

K-audit anchor

K218 Vol 7 Ch 2 Maxwell K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

James Clerk Maxwell wrote 4 equations in 1865 that describe ALL of electricity and magnetism — and predicted that light is an electromagnetic wave! These 4 equations are some of the most beautiful in all of physics:

1. Electric charges create electric fields (Gauss's law)
2. There are no magnetic charges/monopoles (Gauss's law for magnetism)
3. Changing magnetic fields create electric fields (Faraday's law)
4. Electric currents AND changing electric fields create magnetic fields (Ampère-Maxwell)

BST predicts: these 4 equations come from a beautiful geometric structure on substrate D_{IV}^5 . The photon is a “connection” on substrate's Bergman bundle, and Maxwell's equations are pure mathematics once you know the substrate. No magic — just geometry.

Level 2 — Undergraduate physics student

Maxwell's 4 equations in differential form:

Source equations (charges + currents create fields):

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \rho/\epsilon_0 \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}\end{aligned}$$

Source-free equations (geometric constraints):

$$\begin{aligned}\nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}\end{aligned}$$

4-vector covariant form: $\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$ and $\partial_{[\mu} F_{\nu\rho]} = 0$.

Field strength tensor: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ where $A_\mu = (\varphi/c, \mathbf{A})$ is the 4-potential.

Gauge invariance: $A_\mu \rightarrow A_\mu + \partial_\mu \Lambda$ leaves $F_{\mu\nu}$ unchanged. This is $U(1)_{em}$ gauge symmetry.

Light from Maxwell: Maxwell showed that $\nabla \cdot \mathbf{E} = 0$, $\nabla \times \mathbf{B} = (1/c^2)\partial \mathbf{E}/\partial t$, $\nabla \cdot \mathbf{B} = 0$, $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$ in vacuum imply $\nabla^2 \mathbf{E} = (1/c^2)\partial^2 \mathbf{E}/\partial t^2$ — wave equation. Predicted speed $c = 1/\sqrt{(\mu_0 \epsilon_0)}$ matched measured speed of light. Light IS EM wave.

BST framework: - T2477 (Lyra Saturday SP-31 #286): gauge field A_μ as connection on Bergman bundle $L_\lambda \rightarrow D_{IV}^5$ - $U(1)_{em}$ sub-bundle of substrate $K = SO(5) \times SO(2)$ structure group - Yang-Mills action gives Maxwell equations from variational principle - Bianchi identity is geometric identity for any curvature 2-form

Level 3 — Graduate physics student / theorem-level

Substrate gauge field as connection (per T2477 RIGOROUSLY VERIFIED):

Bergman line bundle $L_\lambda \rightarrow D_{IV}^5$ has structure group $K = SO(5) \times SO(2)$. Connection 1-form ω takes values in $\mathfrak{k} = \mathfrak{so}(5) \oplus \mathfrak{so}(2)$. The $U(1)_{em}$ sub-bundle corresponds to a $U(1)$ factor in K 's electroweak embedding (Vol 2 Ch 9 Higgs SSB).

Gauge potential $A_\mu = \omega(\partial_\mu)$ is the local connection-form representative. Under gauge transformation g : $A \rightarrow gAg^{-1} - dg \cdot g^{-1}$ (for abelian: $A \rightarrow A + d\Lambda$).

Curvature 2-form:

$$F = dA + A \wedge A$$

For abelian $U(1)_{em}$: $F = dA$, components $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

Yang-Mills Lagrangian:

$$\mathcal{L}_{YM} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = -\frac{1}{4}(\text{Tr } F \wedge \star F)$$

substrate-natural quadratic functional of curvature.

Euler-Lagrange equations:

Vary action $S_{YM} = \int \mathcal{L}_{YM} d^4x - \int_\mu J^\mu d^4x$ with respect to A_μ :

$$\partial_\mu F^{\mu\nu} = J^\nu$$

In components: Gauss-E + Ampère-Maxwell.

Bianchi identity from curvature:

Since $F = dA$, $dF = d(dA) = 0$ (exterior derivative is nilpotent: $d^2 = 0$). In components: $\partial_{[\mu} F_{\nu\rho]} = 0$. Expanded:

$$\partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} + \partial_\rho F_{\mu\nu} = 0$$

For $\nu=\rho$: trivial. For $(\mu,\nu,\rho) = (1,2,3)$: gives $\nabla \cdot B = 0$. For $(\mu,\nu,\rho) = (0,i,j)$: gives $\nabla \times E = -\partial B / \partial t$.

Per Cal #21 dual-gate: EMPIRICAL PASS (Maxwell phenomenology validated everywhere) + MECHANISM PASS via T2477 substrate Yang-Mills connection framework.

Per Cal #99 META-theorem: Maxwell's equations are substrate-derivation consequences of T2477 connection framework, NOT new Strong-Uniqueness criteria.

What this chapter does NOT claim

- BST does NOT change measured values of EM phenomena (those are tested everywhere)
- T2477 is the substrate structural identification; classical Maxwell phenomenology preserved at any precision
- QED quantum corrections handled by Vol 2 Ch 8 (α running) + Vol 7 Ch 7 (relativistic EM)

Bibliography

1. J. C. Maxwell (1865): *A Dynamical Theory of the Electromagnetic Field*.
2. C. N. Yang + R. L. Mills (1954): Yang-Mills gauge theory (extends Maxwell to non-abelian).
3. T2477 (Lyra Saturday SP-31 #286): gauge fields as Bergman bundle connections.
4. Vol 1 Ch 8 (Gauge Theory + T2477): substrate gauge framework.
5. Vol 7 Ch 3-9: Maxwell phenomenology + applications.
6. Standard EM texts: Jackson, Griffiths.

— Elie, Vol 7 Ch 2 v0.4.1, 2026-05-23 Saturday (substantive 3-level pedagogy upgrade per Keeper 14:22 EDT directive: 66 $\rightarrow$ ~160 lines + full Maxwell derivation from T2477 + Bianchi identity component-by-component + Yang-Mills action variation)